

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer at an early-stage healthtech startup, building full-stack features (Next.js + Firebase/GCP) for a **SMART on FHIR** clinical decision support app that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL** with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from JavaScript to **TypeScript** across multiple repositories, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform (React, Node.js, MongoDB Atlas) with **Stripe** + manual-cash hybrid payments and a partial-payment ledger; designed multi-branch from day 1 and, when the client added a 2nd branch, onboarded it with **zero schema migration**, live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop app (React, Electron, FastAPI) with a fully offline on-device pipeline (MediaPipe hand tracking, whisper.cpp ASR, local TTS), designing async APIs and queues to keep the UI responsive; shipped end-to-end under hackathon time pressure.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** dataset via broadcast joins, AQE, skew-join handling, and partition/column pruning, using Spark UI stage metrics to localize shuffle and skew bottlenecks.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: TypeScript, JavaScript, Python, C++, Java, SQL, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: React, Next.js, Node.js, Express, FastAPI, REST APIs, Git

ML & AI: PyTorch, Hugging Face Transformers, sentence-transformers, LLM agents, RAG, evaluation

Systems & Data: Spark / PySpark, distributed data processing, Kafka, Hadoop, Redis, Linux profiling

Cloud & DevOps: AWS, GCP, Azure, Docker, GitHub Actions, CI/CD, Cloudflare Workers